

THEORY AND METHODS
WEIGHTED ESTIMATION OF THE LOCATION
PARAMETER OF A SYMMETRIC STABLE
DISTRIBUTION

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Key words: Location parameter, numerical integration, stable distribution, weighted estimation.

Summary: We propose a class of weighted estimators of the location parameter of a symmetric stable distribution, where the weights decrease exponentially as a function of the scaled absolute deviation of observations from the sample median. The proposed weighted mean outperforms both the unweighted sample mean and the median for stable symmetric distributions with stability index $1 < 2$, and outperforms the optimal trimmed mean for the Cauchy distribution.

1. Introduction

Stable distributions are a class of probability distributions that accommodate skewness and heavy tails. A random variable X is called stable in the broad sense if for X_1 and X_2 independent copies of X and any positive constants a and b , the sum $aX_1 + bX_2$ has the same distribution as $cX + d$ for some $c > 0$ and $d \in \mathbb{R}$. An overview of the properties of stable distributions is given in the books of Nolan (2002) and of Uchaikin and Zolotarev (1999). The summary of the properties of the stable distribution presented in this section is largely taken from Nolan (2002, Chapter 1).

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Stable distributions are characterized by four parameters: an index of stability $\alpha \in (0, 2]$, a skewness parameter $\beta \in [-1; 1]$, a scale parameter $\gamma > 0$ and a location parameter $\delta \in R$. Following Nolan (2002, Section 1.3) we use the notation $S(\alpha, \beta, \gamma, \delta; k)$, where the index k may be added to denote the parameterization of the stable distribution in question. The class of stable distributions contains the normal distribution (with parameters $\alpha = 2$ and $\beta = 0$), the Cauchy distribution (with parameters $\alpha = 1$ and $\beta = 0$) and the Lévy distribution (with parameters $\alpha = \frac{1}{2}$ and $\beta = 1$) as special cases.

In the present paper we focus on symmetric stable distributions $S(\alpha, 0, \gamma, \delta)$, that is, $\beta = 0$. The symmetric and standardized stable distribution $S(\alpha, 0, 1, 0)$, with $\beta = 0$ (symmetry), $\gamma = 1$ (unit scale) and $\delta = 0$ (zero location) we denote as $S(\alpha)$.

For a stable distribution with parameter $\alpha < 2$, $E(|x^r|)$ is finite for $r < \alpha$, but infinite for $r \geq \alpha$. Thus stable distributions with $\alpha < 2$ do not have a finite variance, and for stable distributions with $\alpha \leq 1$ the mean is undefined.

If $f_\alpha(x)$ and $F_\alpha(x)$ respectively denote the density and distribution function of a symmetric stable random variable with $1 \leq \alpha \leq 2$ then the distribution has the following tail behaviour (for large $|x|$):

$$f_\alpha(x) \propto |x|^{-(\alpha+1)}$$

and

$$F_\alpha(x) \propto |x|^{-\alpha}$$

The rate of convergence of the sample mean of a stable distribution is $n^{(\alpha-1)/\alpha}$ (Rachev, 1991). Data from stable distributions within the range 1.5 to 2 are often encountered in financial problems (Rachev & Mittnik, 2000).

It is well known that for distributions with finite second moments, for example a stable distribution with $\alpha = 2$, the (unweighted) sample average outperforms any weighted average as an estimator of location with respect to expected mean squared error (MSE). In contrast, for symmetric stable distributions with $1 \leq \alpha < 2$, that is, distributions with infinite variance, a weighted mean can outperform the unweighted sample mean. In the present paper we propose a class of weighted means for the estimation of the location parameter of a symmetric stable distribution. The weights decrease exponentially as a function of the scaled absolute deviation of the observations from the sample median. A simulation study shows that the proposed weighted mean outperforms the sample average and the sample median for $1 \leq \alpha < 2$. Furthermore, the weighted mean outperforms the optimal trimmed mean for samples from a Cauchy distribution.

2. Weighted means and numerical integration

Estimation of the mean of a distribution can be considered as an approximation of the integral $E(X) = \int x f(x) dx$, where $f(x)$ is the density function of X . In numerical integration an integral is approximated by a weighted sum of the function values, of the form

$$\int g(x) dx \approx \sum_{j=1}^n g(x_j) w_j$$

where x_j and w_j are called the nodes and weights, respectively (see Kreyszig, 1989; Stoer & Bulirsch, 1980). Similarly, the expected value of a random variable X , approximated after observing a sample $x_1, \dots, x_n$ of size n , is given

by

$$\begin{aligned} \int_{-\infty}^{\infty} x f(x) dx &= \int_{-\infty}^{\infty} x dF(x) \\ &\approx \sum_{j=1}^n x_j \frac{1}{n} \\ &= \sum_{j=1}^n x_j w_j \end{aligned}$$

with uniform weights $w_j = 1/n$.

Since $F(X)$ has a uniform distribution on the interval $[0, 1]$ for any distribution function $F(x)$, uniform weights correspond to the expected spacing between consecutive values of order statistics from a uniform distribution for a sample of size $n + 1$, with n spacings. Uniform weights seem reasonable for distributions with relatively light tails, but particularly for distributions with heavy tails larger spacings can be expected between extreme values in the sample. For this reason, most robust estimators of location can be written as weighted averages of the sample values, where the weights follow a step function with small or zero weight for the outlying observations. An example is the trimmed mean.

Stable distributions are distributions with heavy tails. As pointed out above, $P(X > x) \propto |x|^{-\alpha}$ characterizes the tail behaviour of a stable distribution with index α . In the present paper, we investigate weighted averages as estimators of location, with exponentially decreasing weights; the weighted average has a finite MSE.

3. Weighted estimation of the mean of a stable distribution

If $x_1, \dots, x_n$ is a sample of n identically and independently distributed observations from a symmetric stable distribution $S(\alpha, 0, \gamma, \delta)$, we define the weighted estimate of location as

$$\hat{\delta} = \hat{\delta}(\bar{\delta}, \bar{\gamma}, \lambda) = \sum_{j=1}^n x_j w_j$$

where the weights $w_j = w_j(\bar{\delta}, \bar{\gamma}, \lambda)$ are given by

$$w_j = \frac{\exp \left\{ - \left(\frac{|x_j - \bar{\delta}|}{\bar{\gamma}} \right)^\lambda \right\}}{\sum_{j=1}^n \exp \left\{ - \left(\frac{|x_j - \bar{\delta}|}{\bar{\gamma}} \right)^\lambda \right\}} \quad (1)$$

Here $\bar{\delta}$ and $\bar{\gamma}$ are preliminary estimates of the location and scale parameters of the distribution, respectively. The choice of $\bar{\delta}$, $\bar{\gamma}$ and λ as a function of the index of stability α is discussed in Section 4 below where we show that robust estimates of δ and γ perform well when $\alpha < 2$. Thus we choose the sample median $\hat{x}_m$ for $\bar{\delta}$. For $\bar{\gamma}$ we choose Hampel's (1974) robust estimate of scale, namely the multiple of the median of the absolute deviations

$$\hat{\sigma}_H = 1.4826 \cdot \text{med}_j |x_j - \hat{x}_m| \quad (2)$$

Note that there is no instability in the weights (1) because of division by zero or values close to zero when using an exponential form for the weights. Uniform weights are obtained when λ is chosen to be zero.

Under certain conditions it is possible to prove that the weighted mean, where the weights are a function of the observations and of the median, is a consistent estimator of the median, and therefore also of the mean, of symmetric distributions.

Theorem. Let x_m denote the median of a symmetric stable random variable X with index $\alpha > 1$ and density function $f_\alpha(x)$, and let $h \geq 0$ denote a measurable function symmetric about x_m . Assume that h is chosen so that $\text{Var}[h(|X - x_m|)]$ and $\text{Var}[Xh(|X - x_m|)]$ are finite. Let $X_1, \dots, X_n$ be identically and independently distributed random variables with density function f_α . Then the weighted average

$$\hat{X}_w = \frac{\sum_{j=1}^n X_j h(|X_j - x_m|)}{\sum_{j=1}^n h(|X_j - x_m|)} = \frac{(1/n) \sum_{j=1}^n X_j h(|X_j - x_m|)}{(1/n) \sum_{j=1}^n h(|X_j - x_m|)} \quad (3)$$

is a consistent estimator of the median, and therefore also of the mean, of the symmetric stable distribution.

Proof. The absolute deviations about the median are independent, and measurable functions of independent variables are again independent (Uchaikin & Zolotarev, 1999). Thus both the numerator and the denominator in (3) are averages of independent and identically distributed random variables. By the weak law of large numbers it follows that the numerator and denominator of (3) converge in probability respectively to

$$\frac{1}{n} \sum_{j=1}^n h(|X_j - x_m|) \xrightarrow{P} E h(|X - x_m|) \quad \text{as } n \longrightarrow \infty \quad (4)$$

and

$$\frac{1}{n} \sum_{j=1}^n X_j h(|X_j - x_m|) \xrightarrow{P} E [X h(|X - x_m|)] \quad \text{as } n \longrightarrow \infty \quad (5)$$

The symmetry of h and the antisymmetry of $X - x_m$ imply

$$E [(X - x_m) h(|X - x_m|)] = E [X h(|X - x_m|)] - x_m E h(|X - x_m|) = 0$$

so that

$$E [X h(|X - x_m|)] = x_m E h(|X - x_m|) \quad (6)$$

Using Theorem 2.1.3 of Lehmann (1999, Equation (2.1.11) on p. 50), and results (4)–(6) above it follows that

$$\begin{aligned}\hat{X}_w &\xrightarrow{P} \frac{E[Xh(|X - x_m|)]}{Eh(|X - x_m|)} \quad \text{as } n \longrightarrow \infty \\ &= \frac{x_m Eh(|X - x_m|)}{Eh(|X - x_m|)} \\ &= x_m\end{aligned}$$

which proves that $\hat{X}_w$ as in (3) is a consistent estimator of x_m .

In practice the median x_m is unknown. When, in (3), the median x_m is substituted by the sample median $\hat{x}_m$ the result of the theorem still holds, under necessary smoothness of the function h , due to the fact that $\hat{x}_m = x_m + O_p\left(n^{\frac{1}{2}}\right)$.

4. Simulation study

We conducted a simulation study to investigate two questions: Firstly, the weights in Equation (1), or equivalently, the proposed class of weighted estimators, depend on three parameters, namely $\bar{\delta}$, $\bar{\gamma}$ and λ . Thus we attempted to determine which choices of $\bar{\delta}$, $\bar{\gamma}$ and λ in Equation (1) yield an estimator of the location parameter of a symmetric stable distribution with the highest precision. Secondly, we compared the performance of the proposed weighted estimator of location against some alternatives, notably the sample mean, the sample median, and the optimal trimmed mean for samples from the Cauchy distribution. In all simulations (refer to Step 1 of the algorithm below) samples from the $S(\alpha)$ distribution were generated as described by Nolan (2002, Theorem 1.19).

The following algorithm was used to calculate through simulation the RMSE of the weighted estimator $\hat{\delta} = \hat{\delta}(\bar{\delta}, \bar{\gamma}, \lambda)$ for given $\bar{\delta}$, $\bar{\gamma}$ and λ :

Algorithm 1

1. Draw a sample $x_1, \dots, x_n$ of size n from the $S(\alpha)$ distribution.
2. For a given choice of $\bar{\delta}, \bar{\gamma}$ and λ , calculate the weighted estimate $\hat{\delta} = \hat{\delta}(\bar{\delta}, \bar{\gamma}, \lambda) = \sum_{j=1}^n x_j \cdot w_j(\bar{\delta}, \bar{\gamma}, \lambda)$, where the weights w_j are given in Equation (1).
3. Repeat Steps 1 and 2 $N = 10^6$ times, yielding N realizations $\hat{\delta}_1, \dots, \hat{\delta}_N$ of $\hat{\delta}$.
4. Calculate the root mean squared error (RMSE) of $\hat{\delta}$ as

$$\text{RMSE}(\hat{\delta}) = \sqrt{\frac{1}{N} \left(\sum_{i=1}^N \hat{\delta}_i^2 \right)}$$

4.1 Choice of $\bar{\delta}$

We compared the sample mean $\bar{x} = (1/n) \sum_{i=1}^n x_i$ to the sample median $\hat{x}_m$ as choices for $\bar{\delta}$ in Equation (1). Specifically, we investigated which of the two choices for $\bar{\delta}, \bar{x}$ or $\hat{x}_m$, yielded the weighted estimator with smallest RMSE.

Using Algorithm 1 we calculated the RMSE of two weighted estimators, $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ and $\hat{\delta}(\bar{x}, \hat{\sigma}_H, \lambda)$. In all simulations $\bar{\gamma}$ was chosen to be $\hat{\sigma}_H$, a choice motivated by the results of the simulations described in Section 4.2 below.

Table 1 presents the RMSE of $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ and of $\hat{\delta}(\bar{x}, \hat{\sigma}_H, \lambda)$ for a sample size of $n = 20$, for $1 \leq \alpha \leq 2$ and for two choices of λ .

For $\lambda = 1$, $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ has smaller RMSE than $\hat{\delta}(\bar{x}, \hat{\sigma}_H, \lambda)$ when $\alpha \leq 1.7$, the two estimators have approximately equal RMSE when $\alpha = 1.8$, and $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ has slightly higher RMSE than $\hat{\delta}(\bar{x}, \hat{\sigma}_H, \lambda)$ when $1.9 \leq \alpha \leq 2$. For $\lambda = 0.4$, which is a better choice for λ when α is close to 2 (see Section 4.3 below), $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ has smaller RMSE than $\hat{\delta}(\bar{x}, \hat{\sigma}_H, \lambda)$ when $\alpha = 1.8$, and the two estimators have approximately equal RMSE when $\alpha = 1.9$. In all

cases, when $\alpha \leq 1.5$, the difference in RMSE can be quite drastically in favour of $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$.

Overall, and in particular when $\alpha = 1.9$, or when α is uncertain, the sample median $\hat{x}_m$ seems an effective and robust choice for $\bar{\delta}$. Results of simulations for other sample sizes ($n = 50$, $n = 100$; not shown here) are very similar to the results presented in Table 1.

4.2 Choice of $\bar{\gamma}$

Similar to the investigation described in Section 4.1, we compared the following three estimates of scale as choices for $\bar{\gamma}$ in Equation (1):

- The sample standard deviation, $\hat{\sigma}$.
- Hampel's robust estimate of scale, $\hat{\sigma}_H$, as given in Equation (2).
- Croux and Rousseeuw's (1992) robust estimate of scale, $\hat{\sigma}_{CR}$, given by

$$\hat{\sigma}_{CR} = 1.1926 \cdot \text{med}_i (\text{med}_j (|x_j - x_i|)) \quad (7)$$

As before, we calculated the RMSE of the resulting three weighted estimators, namely $\hat{\delta}(x_m, \hat{\sigma}, \lambda)$, $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ and $\hat{\delta}(x_m, \hat{\sigma}_{CR}, \lambda)$. In all simulations $\bar{\delta}$ was chosen to be the sample median $\hat{x}_m$, a choice motivated by the results of the simulations described in Section 4.1 above.

Table 2 presents the RMSE of $\hat{\delta}(x_m, \hat{\sigma}, \lambda)$, $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ and $\hat{\delta}(x_m, \hat{\sigma}_{CR}, \lambda)$, for a sample size of $n = 20$, for $1 \leq \alpha \leq 2$ and for two choices of λ .

For both $\lambda = 1$ and $\lambda = 0.4$, $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ and $\hat{\delta}(x_m, \hat{\sigma}_{CR}, \lambda)$ have practically identical RMSE for all α . Since $\hat{\sigma}_H$ is easier to calculate than $\hat{\sigma}_{CR}$, only $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ will be discussed further.

For $\lambda = 1$, $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ has smaller RMSE than $\hat{\delta}(x_m, \hat{\sigma}, \lambda)$ when $\alpha \leq 1.5$ but $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ has slightly higher RMSE than $\hat{\delta}(x_m, \hat{\sigma}, \lambda)$ when $1.6 \leq \alpha \leq 2$. For $\lambda = 0.4$, which is a better choice for λ when α is close to

2 (see Section 4.3 below), $\widehat{\delta}(x_m, \widehat{\sigma}_H, \lambda)$ has smaller RMSE than $\widehat{\delta}(x_m, \widehat{\sigma}, \lambda)$ when $\alpha \leq 1.9$, and the two estimators have approximately equal RMSE when $\alpha = 2$. When α is close to 1 the difference in RMSE can be quite drastically in favour of $\widehat{\delta}(x_m, \widehat{\sigma}_H, \lambda)$.

As before, results of simulations for other sample sizes ($n = 50$, $n = 100$; not shown here) are very similar to the results presented in Table 2.

4.3 Choice of λ as function of α

Using Algorithm 1, we calculated the RMSE of $\widehat{\delta} = \widehat{\delta}(\widehat{x}_m, \widehat{\sigma}_H, \lambda)$ for all combinations of $\alpha = 1$ to $\alpha = 2$ in steps of 0.1 and $\lambda = 0$ to $\lambda = 3$ in steps of 0.1, for sample sizes $n = 20$, $n = 50$ and $n = 100$. Thus, in all simulations $\bar{\delta}$ was chosen to be the sample median $\widehat{x}_m$ and $\bar{\gamma}$ was chosen to be $\widehat{\sigma}_H$, choices motivated by the results of the simulations described in Sections 4.1 and 4.2 above. For each given value of α and n , we determined that value of λ for which the RMSE of $\widehat{\delta} = \widehat{\delta}(\widehat{x}_m, \widehat{\sigma}_H, \lambda)$ is minimized.

In Figure 1 we plot the λ minimizing the RMSE of $\widehat{\delta} = \widehat{\delta}(\widehat{x}_m, \widehat{\sigma}_H, \lambda)$ as a function of α , for all three sample sizes investigated. The figure shows that the optimal λ as a function of α is practically independent of sample size: for $1.4 \leq \alpha \leq 2$ the optimal λ is in fact identical for all three sample sizes, and for $1 \leq \alpha \leq 1.3$ the optimal differs by at most 0.1 between the three sample sizes. The detailed simulation results show that differences in RSME associated with a change in λ of 0.1 are in the order of only 0.1%.

The optimal λ is zero (implying uniform weights) for $\alpha = 2$ (Normal distribution), as is well known. As α decreases, the optimum λ increases, reaching a value of $\lambda \approx 1.3$ for $\alpha = 1$ (Cauchy distribution). However, the detailed simulation results show that the RMSE of $\widehat{\delta} = \widehat{\delta}(\widehat{x}_m, \widehat{\sigma}_H, \lambda)$ is rather insensitive to the choice of λ when α is close to 1, so that a choice of $\lambda = 1$ is in fact close to optimal for all $\alpha < 1.5$ (when $\alpha = 1$ the RMSE for $\lambda = 1$ is

only about 0.5% higher than the RMSE for $\lambda = 1.3$). When $\alpha \geq 1.5$, a choice of $\lambda = 1$ might not be optimal, but will certainly yield a robust estimator since $\hat{\delta} = \hat{\delta}(\hat{x}_m, \hat{\sigma}_H, \lambda)$ is increasingly robust with increasing λ . Thus $\lambda = 1$ seems a good overall choice, optimal or near optimal when $\alpha < 1.5$, and robust when $\alpha \geq 1.5$.

When the assumption is justified that $1.5 \leq \alpha < 2$, as is typically the case for financial data, λ might be chosen around 0.5: on the high side, say $\lambda = 0.6$, if a robust estimator is preferred, or $\lambda = 0.4$ when α might be reasonably close to 2 and the weighted estimator should perform well compared to uniform weights (when $\alpha = 2$).

4.4 Performance of weighted estimate relative to alternatives

We compared the performance of five estimators of location of symmetric stable distributions:

- The (unweighted) sample mean
- The sample median
- The weighted mean $\hat{\delta}(\hat{x}_m, \hat{\sigma}_H, 1)$
- The weighted mean $\hat{\delta}(\hat{x}_m, \hat{\sigma}_H, 0.4)$
- The trimmed mean using the central 24% of the data, which is the optimal trimmed mean for samples from the Cauchy distribution (Rothenberg, Fisher & Tilanus, 1964; Fama & Roll, 1968).

The RMSE of these five estimators was calculated through simulation, as described above. In Figure 2 the RMSE of the estimators (excluding the sample mean) is plotted against α for a sample size of $n = 20$.

For $\alpha = 2$ (Normal distribution), the sample mean has a RSME of 0.32, which as expected is the lowest of all five estimators. However, for $\alpha = 1.9$ the RSME of the sample mean is 0.43 and increases so rapidly as approaches 1 (for example, RMSE = 1.68 for $\alpha = 1.7$) that the RSME curve could not

be included in Figure 2. The sample mean clearly is not competitive when $\alpha \leq 1.9$.

When $\alpha < 1.5$ the weighted mean with $\lambda = 1$ performs best of the five estimators investigated. Specifically, for $\alpha = 1$ (Cauchy distribution) this weighted mean outperforms the optimal trimmed mean. Furthermore, the weighted mean outperforms the median for all $1 \leq \alpha \leq 1.9$.

The weighted mean with $\lambda = 0.4$ outperforms both the median and the weighted mean with $\lambda = 1$ when $\alpha > 1.5$. Thus the weighted mean with $\lambda = 0.4$ seems a good choice for the practically important case when data come from a stable distribution with stability index $\alpha > 1.5$.

The simulation results for samples sizes of $n = 50$ and $n = 100$ were similar.

5. Discussion

In this paper we have proposed a class of weighted estimators of the location parameter of a symmetric stable distribution. The weights depend on preliminary estimates of the location and scale of the distribution, and on a power parameter λ (see Equation (1)).

Through a simulation study we have shown that simple robust estimates of location (the sample median) and of scale (Hampel's robust scale) lead to optimal or near optimal weights under a wide range of conditions. Depending on the stability index α of the distribution, $\lambda = 1$ seems a good overall choice for the power parameter, which is optimal or near optimal when the stability index of the distribution is low ($\alpha < 1.5$), and is a robust choice when $\alpha = 1.5$. In many practical applications, such as in the analysis of financial data, the assumption is justified that $1.5 \leq \alpha < 2$. In that case, a choice of $\lambda = 0.4$ is

optimal or near optimal, and the associated weighted estimator performs well even compared to the sample mean when $\alpha = 2$.

Simulation results show furthermore that, with appropriate choices of $\bar{\delta}$, $\bar{\gamma}$ and λ , the proposed weighted mean outperforms both the sample mean and the sample median for stable symmetric distributions with stability index $1 \leq \alpha < 2$. Generally, the RMSE of the proposed weighted mean is approximately 5–10% lower than the RMSE of the sample median. The sample mean is not competitive when $\alpha \leq 1.9$. Finally, the weighted mean outperforms the optimal trimmed mean for the Cauchy distribution ($\alpha = 1$), having approximately 5% lower RMSE when $n = 20$.

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Tables

Table 1 Choice of $\bar{\delta}$: RMSE of $\hat{\delta}(\bar{x}, \hat{\sigma}_H, \lambda)$ and $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ ($n = 20$)

α	Root Mean Squared Error			
	$\lambda = 1$		$\lambda = 0.4$	
	$\hat{\delta}(\bar{x}, \hat{\sigma}_H, 1)$	$\hat{\delta}(x_m, \hat{\sigma}_H, 1)$	$\hat{\delta}(\bar{x}, \hat{\sigma}_H, 0.4)$	$\hat{\delta}(x_m, \hat{\sigma}_H, 0.4)$
1.0	6.264	0.351	4.304	0.438
1.1	3.191	0.356	1.471	0.418
1.2	1.978	0.359	0.859	0.402
1.3	1.214	0.362	0.572	0.390
1.4	0.858	0.366	0.490	0.379
1.5	0.602	0.368	0.443	0.370
1.6	0.495	0.371	0.405	0.362
1.7	0.410	0.372	0.377	0.354
1.8	0.371	0.374	0.355	0.348
1.9	0.346	0.375	0.336	0.342
2.0	0.330	0.376	0.320	0.337

Notes: $\bar{x}$: sample mean; x_m : sample median
 $\hat{\sigma}_H$: Hampel's robust scale estimate

Table 2 Choice of $\bar{\gamma}$: RMSE of $\hat{\delta}(\hat{x}_m, \hat{\sigma}, \lambda)$, $\hat{\delta}(x_m, \hat{\sigma}_H, \lambda)$ and $\hat{\delta}(x_m, \hat{\sigma}_{CR}, \lambda)$ ($n = 20$)

α	Root Mean Squared Error					
	$\lambda = 1$			$\lambda = 0.4$		
	$\hat{\delta}(\hat{x}_m, \hat{\sigma}, 1)$	$\hat{\delta}(x_m, \hat{\sigma}_H, 1)$	$\hat{\delta}(x_m, \hat{\sigma}_{CR}, 1)$	$\hat{\delta}(\hat{x}_m, \hat{\sigma}, 0.4)$	$\hat{\delta}(x_m, \hat{\sigma}_H, 0.4)$	$\hat{\delta}(x_m, \hat{\sigma}_{CR}, 0.4)$
1.0	8.485	0.351	0.352	157.860	0.437	0.445
1.1	2.517	0.355	0.354	57.197	0.419	0.424
1.2	0.772	0.359	0.358	57.212	0.403	0.407
1.3	0.421	0.362	0.361	6.233	0.391	0.393
1.4	0.384	0.365	0.364	1.294	0.379	0.381
1.5	0.379	0.368	0.367	0.899	0.369	0.370
1.6	0.365	0.371	0.369	0.489	0.362	0.362
1.7	0.364	0.372	0.371	0.404	0.355	0.355
1.8	0.366	0.374	0.373	0.363	0.348	0.348
1.9	0.369	0.375	0.374	0.347	0.342	0.342
2.0	0.372	0.375	0.375	0.335	0.337	0.337

Notes: $\hat{x}_m$: sample median; $\hat{\sigma}$: sample standard deviation; $\hat{\sigma}_H$: Hampel's robust estimate
 $\hat{\sigma}_{CR}$: Croux and Rousseeuw's robust scale estimate

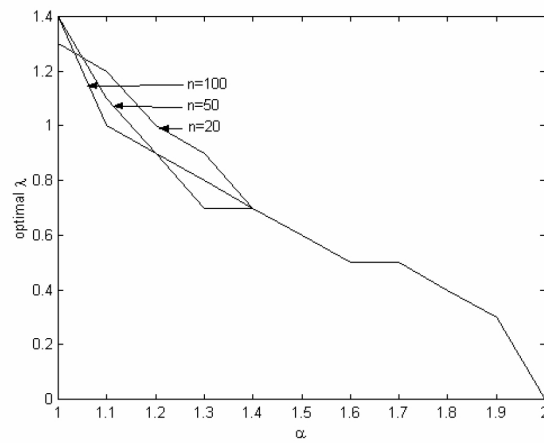


Figure 1 Optimal λ as a function of α ($n = 20, 50$ and 100)

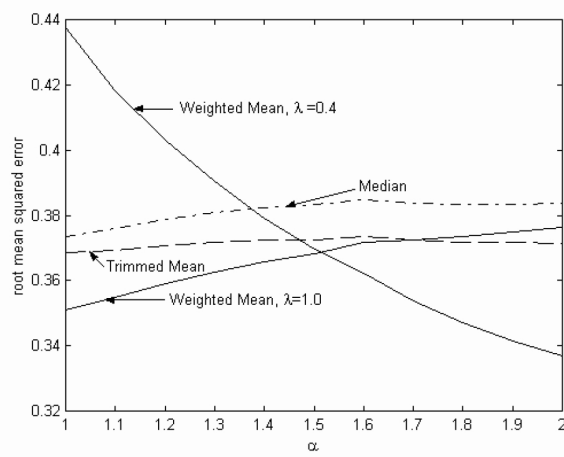


Figure 2 RMSE of four estimators of location ($n = 20$)